SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload
Student Name	J. Thushany
Student ID	JF24ICT09
Implementation Status	Completed – Testing in Progress
Date	12 October 2026

1. Change Request Summary

1.1 Business Requirement

The system required a profile image upload facility to improve student identification and provide a more user-friendly student management interface. Administrators needed the ability to upload an optional profile image when creating a student account. The uploaded image should be stored on the server and displayed in the administrator's student list and the student's own profile dashboard.

1.2 Existing Behavior

Before implementation:

- Administrators could create student accounts by entering student details.
- Student records displayed registration number, name, email address, and phone number.
- The student dashboard displayed initials instead of an uploaded profile photograph.
- No profile image upload facility was available in the student creation form.
- Student profile information was retrieved through the existing student API.

1.3 Requested Change

The following functionality was required:

- Provide an optional profile image field in the Add New Student form.

- Accept PNG, JPG, and JPEG image formats only.
- Restrict uploaded image size to a maximum of 2 MB.
- Reject unsupported file types and oversized files.
- Store uploaded images in a server-side upload directory.
- Save the uploaded image path in the existing students.profile_photo database column.
- Display the uploaded image in the administrator's student list.
- Display the uploaded image on the student's profile dashboard.
- Display student initials when no image is available or the image cannot be loaded.
- Preserve existing student creation, editing, deletion, authentication, and authorization functionality.

2. Implementation

2.1 Implementation Summary

CR-010 was implemented to support optional student profile image uploads. The administrator can select an image while creating a student account. The frontend validates the selected file type and size before submitting the student details.

The backend uses Multer to process uploaded images, checks the permitted image extensions and MIME types, verifies the PNG or JPEG file signature, and stores the image in the server-side uploads directory. The image path is saved in the existing profile_photo column.

The administrator dashboard and student dashboard were updated to display profile images. When an image is unavailable, the interface displays the student's initials as a fallback.

2.2 Affected Files

File	Change	Purpose
backend/src/app.js	Modified / Verified	Serves uploaded images through the /uploads static route.
backend/src/routes/studentRoutes.js	Modified	Integrates image upload middleware with the student creation endpoint.

backend/src/middleware/uploadMiddleware.js	Modified	Validates image type and size, verifies file signatures, and stores uploaded images.
backend/src/controllers/studentController.js	Modified	Processes the uploaded image path and includes it in student creation and profile responses.
backend/src/models/studentModel.js	Modified	Saves and retrieves the profile_photo field.
frontend/src/services/api.js	Modified	Supports multipart FormData requests while preserving authentication headers.
frontend/src/components/StudentFormModal.jsx	Modified	Adds image selection, preview, frontend validation, and image upload during student creation.
frontend/src/pages/DashboardPage.jsx	Modified	Displays student profile images in the administrator's student list, with initials as a fallback.
frontend/src/pages/StudentDashboardPage.jsx	Modified	Displays the student's uploaded profile image on the student dashboard.
frontend/src/styles/dashboard.css	Modified	Styles profile images and fallback initials in the administrator dashboard.
frontend/src/styles/studentDashboard.css	Modified	Styles profile images and fallback initials in the student dashboard.
Database	Existing column used	Uses the existing students.profile_photo column; no additional column is required.

2.3 Technical Changes

Profile Image Upload

An optional file input was added to the Add New Student form. The input accepts PNG, JPG, and JPEG images and supports a maximum file size of 2 MB.

Frontend Validation

The frontend checks the selected file's extension, MIME type, and size before submission. Student details and the selected image are submitted using FormData.

Backend Validation and Storage

Multer is used to process the uploaded image. The backend checks the allowed extensions and MIME types, verifies the PNG or JPEG signature, limits the upload to one file, and restricts the file size to 2 MB.

Uploaded images are stored in the backend uploads directory with unique filenames to reduce filename collisions.

Database Integration

The uploaded image path is stored in the existing students.profile_photo column. Student queries retrieve this field so the frontend can access the image location.

Image Display and Fallback

The administrator dashboard displays a profile image for each student when one is available. The student dashboard also displays the uploaded image. Initials are shown when no image is stored or the image fails to load.

Existing Functionality Preservation

The implementation is intended to preserve existing student creation, editing, deletion, authentication, authorization, and student profile retrieval. These functions must be regression-tested before final acceptance.

2.4 Implementation Logic

1. The administrator opens the Add New Student form.
2. The administrator enters the student's details and optionally selects a profile image.
3. The frontend validates the selected image format and file size.
4. The student details and image are submitted to the backend using FormData.

5. The upload middleware validates the file and checks its signature.
6. The backend stores the uploaded image in the uploads directory.
7. The image path is saved in the students.profile_photo column when the student record is created.
8. The student API returns the profile image path with the student information.
9. The administrator dashboard displays the uploaded image or fallback initials.
10. The student dashboard displays the uploaded image or fallback initials.

2.5 Error and Data Handling

The implementation includes the following error-handling measures:

- Unsupported file formats are rejected.
- Files exceeding 2 MB are rejected.
- Multiple image uploads are not permitted in a single upload request.
- Invalid PNG or JPEG file signatures are rejected.
- A missing profile image is handled by displaying student initials.
- An image loading failure triggers the fallback initials display.
- Existing API error handling is retained for student operations.

The image signature check verifies the expected file header. It is a basic file validation measure and does not constitute complete image decoding or comprehensive malware scanning.

3. Testing & Verification

3.1 Testing Approach

The feature should be verified through manual frontend and backend testing.

Testing Types:

- Functional testing
- Positive testing
- Negative testing
- File validation testing
- UI testing

- Image storage and retrieval testing
- Error-handling testing
- Regression testing

Environment: Development

Browser: Google Chrome

Test Role: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the Add New Student form	Optional Profile Image field is displayed.	To be recorded	Pending
TC-02	Upload a valid PNG image	Image is accepted and stored successfully.	To be recorded	Pending
TC-03	Upload a valid JPG/JPEG image	Image is accepted and stored successfully.	To be recorded	Pending
TC-04	Submit the form without an image	Student is created without a profile image.	To be recorded	Pending
TC-05	Upload a PDF or unsupported file	File is rejected with a suitable error message.	To be recorded	Pending
TC-06	Upload an image larger than 2 MB	Upload is rejected with a size error message.	To be recorded	Pending
TC-07	Upload a file with an image extension but an invalid signature	File is rejected by backend validation.	To be recorded	Pending
TC-08	View a student with an uploaded image in the admin dashboard	The uploaded profile image is displayed.	To be recorded	Pending
TC-09	View the student dashboard after uploading an image	The student's profile image is displayed.	To be recorded	Pending
TC-10	View a student without an image or with an unavailable image	Student initials are displayed as a fallback.	To be recorded	Pending

TC-11	Verify existing student CRUD operations	Existing student operations continue to work.	To be recorded	Pending
TC-12	Verify existing authentication and authorization	Existing access restrictions remain effective.	To be recorded	Pending

3.3 Test Summary

Test Result	**Count**
Total Test Cases	12
Passed	To be recorded
Failed	To be recorded
Pending	12 until testing is completed
Pass Rate	To be calculated after testing

Overall Test Result: Pending verification

4. Acceptance Criteria Verification

ID	**Acceptance Criterion**	**Verification Method**	**Status**
AC-001	Administrators can select an optional profile image when creating a student.	Add New Student form test	Implemented; verify
AC-002	PNG, JPG, and JPEG images are accepted.	Valid file upload tests	Implemented; verify
AC-003	Unsupported file types are rejected.	Negative file upload test	Implemented; verify
AC-004	Uploaded files are limited to 2 MB.	File size validation test	Implemented; verify
AC-005	Uploaded images are stored on the server and their paths are saved in the database.	Upload directory and database verification	Implemented; verify
AC-006	The administrator dashboard displays profile images.	Admin dashboard test	Implemented; verify

AC-007	The student dashboard displays the student's profile image.	Student dashboard test	Implemented; verify
AC-008	Initials appear when an image is unavailable.	Fallback UI test	Implemented; verify
AC-009	Existing student CRUD functionality remains operational.	Regression testing	Pending verification
AC-010	Existing authentication and authorization are preserved.	Access control testing	Pending verification

5. Evidence

Evidence 01 – Admin Student Creation and Profile Image

Related: AC-001, AC-002, AC-003, AC-004

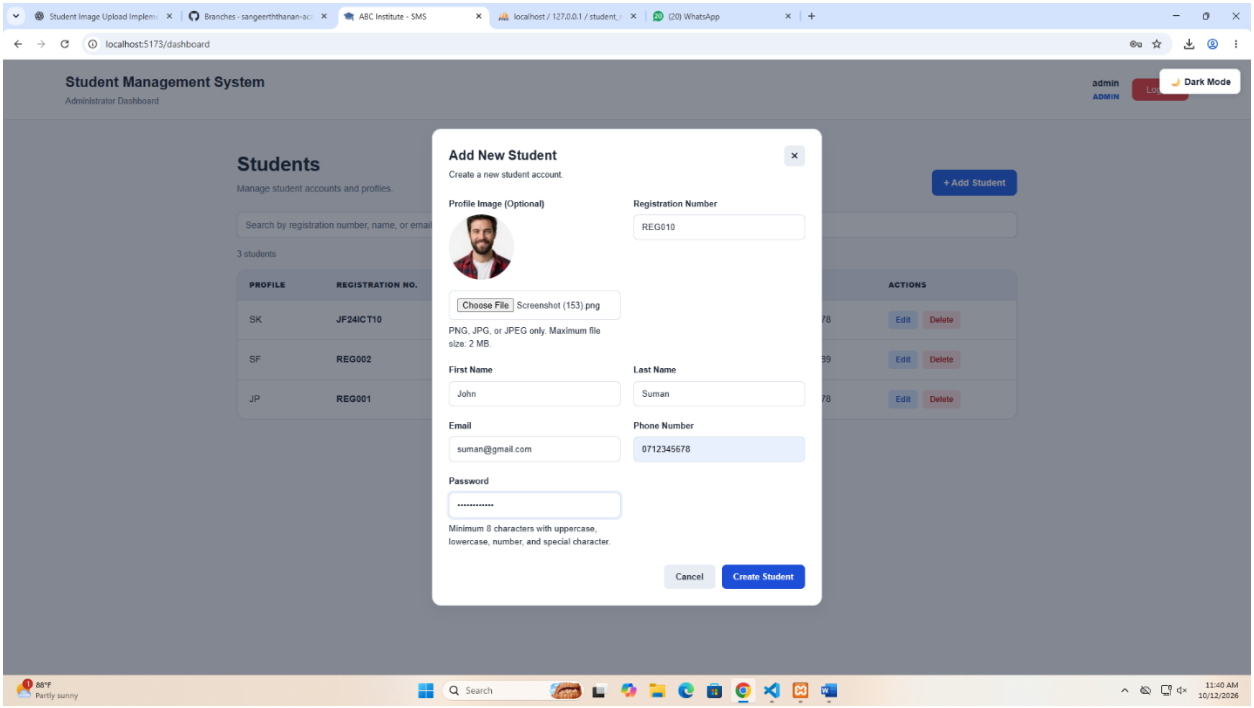


Figure 1: Administrator's Add New Student form showing the optional Profile Image field and a successfully created student with a profile image in the student list.

Evidence 02 – Student Profile Image Display

Related: AC-005, AC-006, AC-007, AC-008

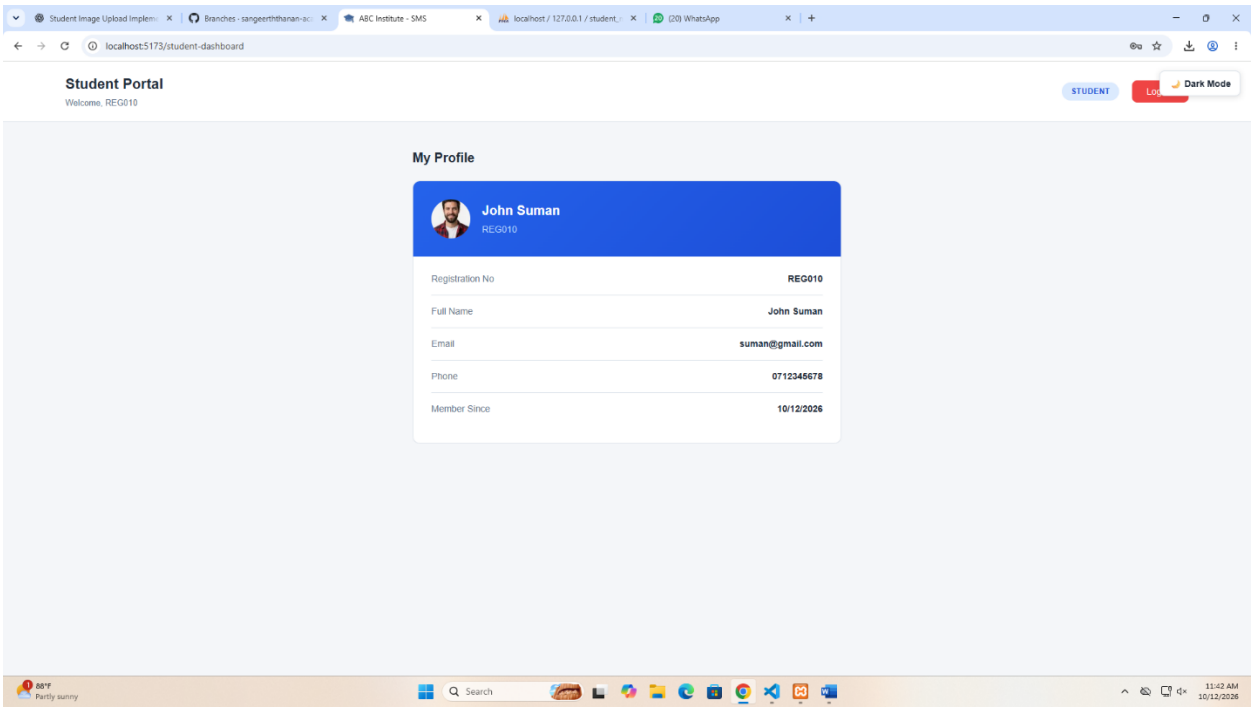


Figure 2: Student dashboard showing the uploaded profile image alongside the student's registration number and profile details. If applicable, include the initials fallback for a student without an image.

6. Final Result

Item	Result
Requirements	Implemented; verification in progress
Implementation	Completed
Testing	Pending
Acceptance Criteria	Pending final verification
Existing CRUD	Pending regression verification
Final Status	Pending testing and acceptance

6.1 Final Summary

CR-010 introduces an optional student profile image upload facility to the Student Management System. The administrator can select an image when creating a student account. The implementation supports PNG, JPG, and JPEG formats, enforces a 2 MB upload limit, performs basic backend file validation, stores uploaded images on the server, and saves the image path in the existing students.profile_photo database column.

The administrator dashboard and student dashboard have been updated to display uploaded profile images, with initials used as a fallback when an image is unavailable. Final acceptance depends on completing the defined upload, validation, display, and regression tests.

6.2 Issues / Limitations

- Final functional and regression testing must be completed before marking all test cases as passed.
- Profile images are stored on the application server; backup and deployment configuration should account for the uploads directory.
- The current validation checks the file extension, MIME type, and image signature. More comprehensive image decoding and security scanning would be appropriate for a production deployment.
- Updating an existing student's profile image is not included unless separately implemented.

Student Name: J. Thushany

Student ID: JF24ICT09

Date: 12 October 2026

Signature: J.Thushany